MAJOR PROJECT - II

- **Title –** AI-Powered Video Knowledge Mapper
- **Team Members:**
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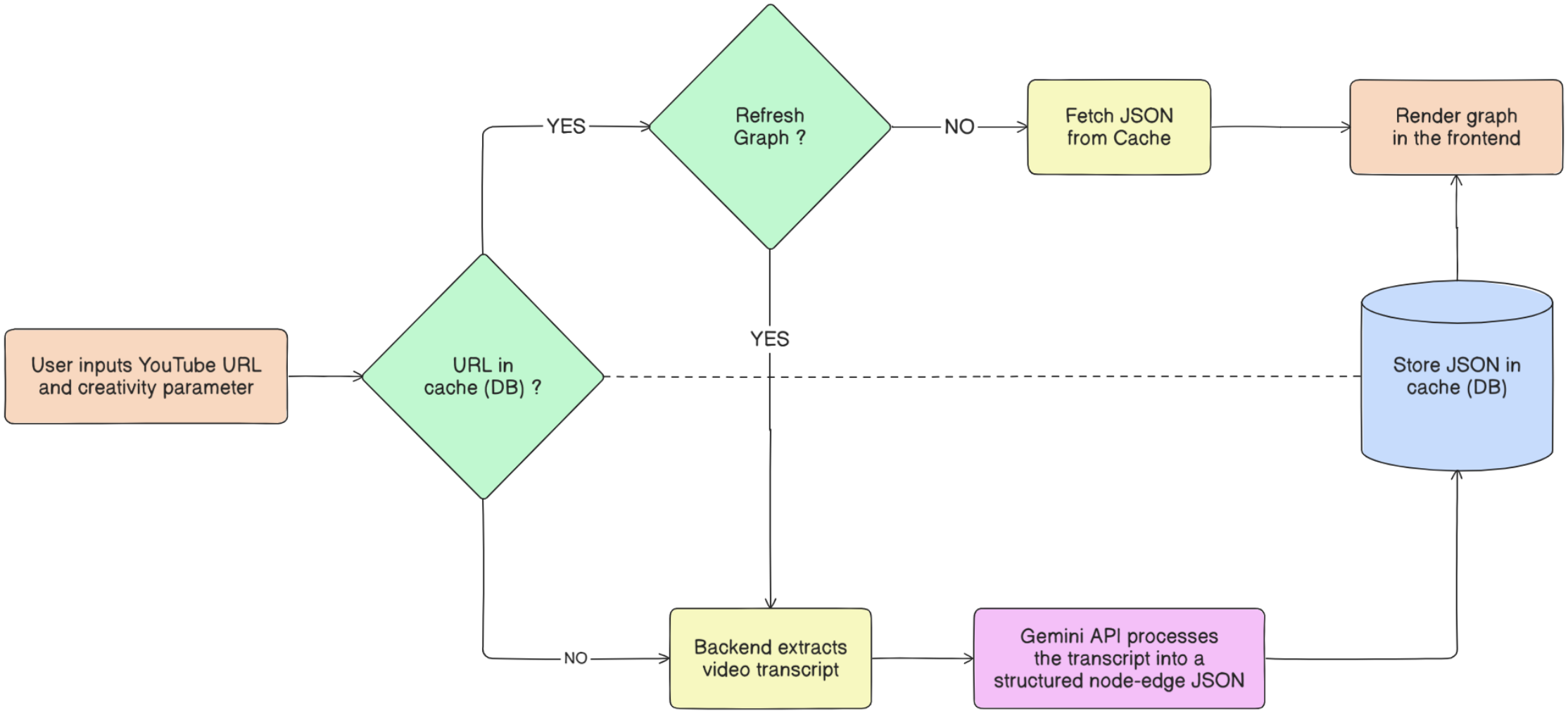
OBJECTIVES OF THE PROJECT

The following points explain the objectives this project will try to achieve:

- **Develop an AI-Based Knowledge Mapping System:** Build a platform capable of extracting video transcripts and converting them into structured mind maps automatically.
- **Implement Generative AI Integration:** Utilize modern generative AI APIs to interpret transcripts and generate hierarchical concept structures in structured formats.
- **Enable Interactive Visualization:** Create an intuitive frontend that displays generated mind maps as interactive node-edge diagrams.
- **Enhance Technical and Analytical Skills:** Provide hands-on experience in full-stack AI development, structured output design, and frontend visualization technologies.

METHODOLOGY



HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements

Field	Technology
CPU	Core i5 or better
GPU	Integrated or Discrete
RAM	8 GB minimum
Storage	10 GB available space
Internet	Stable broadband

Software Requirements

Field	Technology
Frontend	React.js (JavaScript)
Backend	FastAPI (Python)
Database	PostgreSQL
Visualization	ReactFlow, Dagre
GenAI API	Google Gemini API (Gemini 2.0 Flash)

RATIONALE OF THE PROJECT

Traditional video-based learning forces users into a linear consumption model, which creates several inefficiencies:

- **Time Inefficiency:** Users waste valuable time scrubbing through timestamps just to locate specific concepts or definitions.
- **Lack of Structural Overview:** Long-form technical videos fail to provide a top-down, holistic view of how different sub-topics interconnect.
- **High Cognitive Load:** Manually extracting, summarizing, and organizing information from unstructured video content is a tedious barrier to rapid learning.
- **The Core Need:** There is a critical need for an automated system that transforms passive, linear video content into structured, interactive knowledge graphs, enabling faster and non-linear exploration.

EXPECTED OUTCOMES

Impact on Target Audience

- **Accelerated Learning:** Drastically reduces the time required to grasp complex subjects by replacing hours of video watching with scannable, visual graphs.
- **Enhanced Comprehension:** Simplifies technical topics by providing a clear overview of interconnected ideas.
- **Active Engagement:** Transforms passive video consumption into interactive visual graphs.

Benefits of the Solution

- **Automated Knowledge Extraction:** Eliminates the manual, tedious effort of taking notes of long video-based content.
- **Resource Optimization:** Smart PostgreSQL database caching minimizes GenAI API calls, reducing operational costs and ensuring lightning-fast response times.
- **Scalable Information Format:** Converts unstructured video transcripts into a highly standardized JSON format.